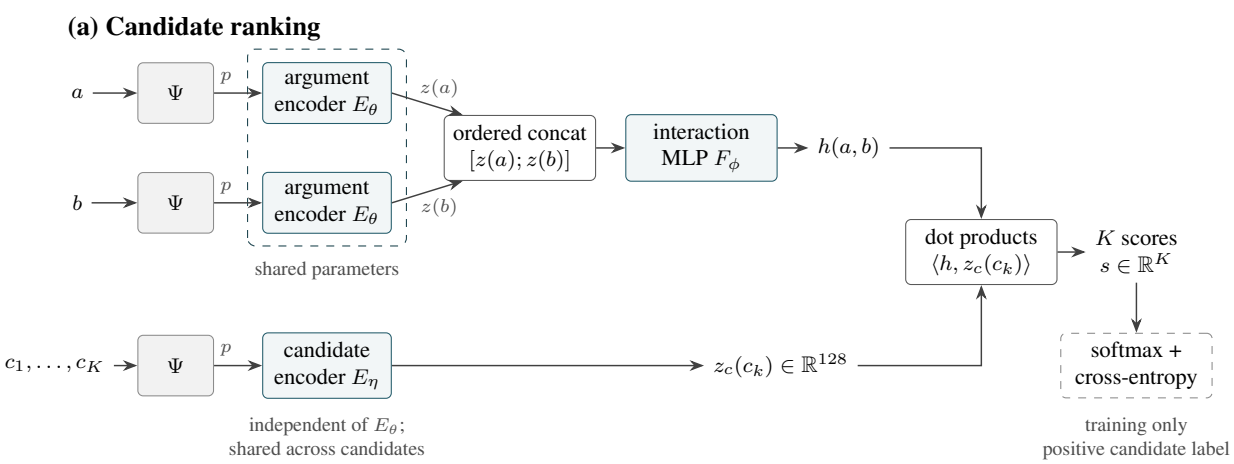

(a) Candidate ranking
a
Ψ
p
argument encoder E_θ
z(a)
b
Ψ
p
argument encoder E_θ
z(b)
shared parameters
ordered concat [z(a); z(b)]
interaction MLP F_ϕ
h(a, b)
dot products ⟨h, z_c(c_k)⟩
K scores s ∈ ℝ^K
softmax + cross-entropy
training only positive candidate label
c_1, . . . , c_K
Ψ
p
candidate encoder E_η
z_c(c_k) ∈ ℝ^128
independent of E_θ; shared across candidates
(b) Layers inside each learned module
Argument encoder E_θ
affine: p → 256
GELU
affine: 256 → 128
Interaction MLP F_ϕ
affine: 256 → 512
GELU
affine: 512 → 128
Candidate encoder E_η
affine: p → 256
GELU
affine: 256 → 128